

# 1 Appendix-I

*All of these interviews have been summarised.*

## 2 Industry Expert Validation Interviews

This appendix documents the semi-structured validation interviews conducted with domain experts throughout the product development lifecycle. These interactions were designed not merely as retrospective reviews, but as active feedback loops to stress-test our architectural assumptions against the rigorous standards of institutional finance. The insights gathered—ranging from compliance mandates to distributed system topology—were instrumental in refining our design. As detailed in the "Team Update" annotations below, specific expert feedback directly catalyzed architectural pivots (such as the shift to Corrective RAG and the implementation of state persistence), ensuring that the final system is not only theoretically sound but operationally viable for deployment in high-frequency trading and banking environments.

### 2.1 Interview 1: Strategic Validation & Explainability

**Interviewee:** Julia Liu, Head of Finance - fal.ai (Ex-Investment Banker, Goldman Sachs)

**Focus:** Trust, Explainability, and Institutional Adoption

**Q1: Team 76:** "Julia, we are designing this for Tier-1 firms. We are debating whether to let the LLM handle the full trade execution loop to maximize autonomy. However, we are worried about the 'Black Box' problem. In your experience at major banks, is an end-to-end LLM trader viable?"

**Julia Liu:** "From a compliance perspective, an end-to-end LLM trader is a non-starter. Banks do not fear AI being wrong; they fear it being inexplicably wrong. If a system hallucinates a trade, liability becomes impossible to trace. The architecture you are proposing—where the LLM only generates the *hypothesis*, but a deterministic solver (like the Portfolio Agent) strictly calculates allocation based on hard-coded risk limits—is the prudent path. It mirrors the 'Maker-Checker' protocols standard in the industry: one entity proposes, another validates with hard math."

**Q2: Team 76:** "To ensure the system doesn't act on vague instructions, we are forcing users to go through a 'Clarification Agent' first. We are concerned this adds too much friction. Is this level of hand-holding necessary for professionals?"

**Julia Liu:** "Ambiguity is a massive operational risk on a trading desk. If a PM says 'get me exposure to tech,' and the system blindly executes, that is a failure of design. A professional system should stop and ask, 'Do you mean sector exposure or beta exposure?' That isn't annoyance; it is competence. In my experience, semantic guardrails are what separate experimental demos from enterprise-grade tools."

### 2.2 Interview 2: Feature Marketability & Client Needs

**Interviewee:** Charudatt Mulye, Head of Equity and Data Science (India)

**Focus:** FinTech Product Validation, News Sentiment Analysis, and Competitive Edge

**Q1: Team 76:** "Charudatt, given your experience building analytics tools, we know 'Price Alerts' are common. We built the Trigger Agent to fuse price anomalies with real-time 'News Sentiment'. Is this integration actually valuable to a fund manager, or is it superfluous?"

**Charudatt Mulye:** "It is the primary differentiator in modern analytics. Price alerts have become a commodity. What the industry lacks is context. When a stock drops 5%, the immediate question is 'why'. Overlaying real-time news sentiment provides that answer immediately—distinguishing a CEO scandal from a sector rotation. Capturing that qualitative signal and mapping it to volatility in real-time creates the kind of 'Smart Alerting' that adds genuine value to a manager's dashboard."

**Q2: Team 76:** "We are currently ranking sectors purely based on technical momentum (price action). Based on your experience with sophisticated funds, is technical analysis sufficient for a sector rotation tool?"

**Charudatt Mulye:** "A sophisticated fund would likely reject a tool based solely on technicals. In the current market, sector moves are heavily driven by regulation—antitrust laws hitting Tech or tariffs hitting Manufacturing. A robust tool must integrate a layer that checks the regulatory landscape, such as filing analysis, before making a recommendation. Without that fundamental check, a technical breakout signal can often be a trap."

**Team Update:** Following Charudatt's advice on the insufficiency of pure technicals, we paused the Sector Tool release and developed the **Macro Agent with CRAG**. As seen in the final architecture, this agent now cross-references technical signals with a vector database of regulatory filings and tax codes to ensure sector recommendations are robust against regulatory risk.

## 2.3 Interview 3: Hedge Fund Use Case Analysis

**Interviewee:** Rupal Shah, Analyst, Goldman Sachs

**Focus:** Tactical Alpha, Drawdown Control, and Fund Mandates

**Q1: Team 76:** "Rupal, we designed this specifically for Tactical Hedge Funds running Long/Short strategies. These funds need to catch trends early but hate getting 'whipsawed' by noise. We implemented a 'hysteresis' filter (dual-confidence check) so the Trigger Agent only signals on sustained pressure. Is this specific feature valuable for a fund manager?"

**Rupal Shah:** "For a tactical fund, that feature is critical. In Long/Short equity, 'churn' kills returns—if you chase every tick, transaction costs eat your alpha. By enforcing hysteresis, you are essentially coding 'trader discipline' into the bot. It ensures the system captures the meat of the trend without getting chopped up in the noise. That validates that you understand the P&L mechanics of a hedge fund, not just the technology."

**Q2: Team 76:** "We learned that Hedge Funds often have rigid mandates from their Limited Partners regarding leverage and drawdown. Therefore, we decided that while the AI can suggest high-alpha strategies, the *Portfolio Agent's solver* hard-locks the execution within those specific mathematical limits. Is this separation important?"

**Rupal Shah:** "It is the only way a fund manager can sleep at night. In a hedge fund, breaching a risk mandate isn't just a compliance error; it is a fund-ending event. Knowing that the 'Reasoning Layer' (AI) is physically constrained by the 'Calculation Layer' (Solver) allows the manager to trust the system with actual capital. It shifts the product from being just a 'research tool' to a viable 'execution copilot' that respects the fund's survival constraints."

## 2.4 Interview 4: Technical Architecture & Scalability

**Interviewee:** Mihir Karandikar, Member Technical, D.E. Shaw Group

**Focus:** Distributed Systems, Scalability, and State Management

**Q1: Team 76:** "Mihir, we are designing this to scale. We decided to strip all 'memory' out of the domain agents (Market, Macro, etc.) and keep them stateless, putting all state in the Orchestrator. Is this valid for a high-throughput financial system, or should we allow agents to have local memory?"

**Mihir Karandikar:** "No, keep them stateless. That is the standard pattern for high-frequency distributed systems. If you give agents local memory, you can't scale them horizontally because you get state drift. Your 'Stateless Worker / Stateful Orchestrator' design is exactly how we would architect a scalable internal tool at a firm like D.E. Shaw. It allows you to spin up 100 market agents during a volume spike without worrying about synchronizing memory between them."

**Q2: Team 76:** "Currently, we run our high-frequency Market Data ingestion and our heavy-duty SEC Filing processing within the same Pathway computation graph to simplify deployment. Is this monolithic graph structure a risk in production?"

**Mihir Karandikar:** "It is a massive reliability risk. You are architecturally coupling a latency-sensitive stream (Market Ticks) with a compute-intensive stream (PDF Vectorization). If a massive 100-page 10-K filing hits the text pipeline and spikes CPU usage, it creates backpressure that delays your price ticks. In finance, you never let the 'slow path' block the 'fast path.' You must decouple the topologies—run them as separate processes that join asynchronously, ensuring the market data stream remains pristine regardless of document load."

**Team Update:** Validating Mihir's concern on resource contention, we decoupled the architecture. We refactored the system into two distinct Pathway pipelines—one optimized for sub-millisecond market ticks and another for heavy document processing—bridged via a non-blocking lookup join. This ensures that analyzing a complex regulatory document never induces latency in the price alert system, preserving the integrity of the real-time loop.

## 2.5 Interview 5: Quant Execution & Iteration Speed

**Interviewee:** Akshay Tak, Quant Trader, HFT

**Focus:** Strategy Iteration, Backtesting, and Execution Bridge

**Q1: Team 76:** "Akshay, we are building a Strategy Agent that uses Evolutionary Computing to discover trading rules automatically. In the HFT world, is automated hypothesis generation actually useful, or is manual rule-writing still the standard?"

**Akshay Tak:** "Manual rule-writing is the standard largely because automated tools often produce unreadable code. However, utilizing STVGP (Strongly Typed Genetic Programming) changes the conversation. Because it enforces valid logic structures, the output becomes interpretable. For a quant desk, this serves as a massive accelerator. It provides a starting point rather than a blank page, significantly speeding up the R&D cycle for new strategies."

**Q2: Team 76:** "We are putting a lot of effort into using Pathway for streaming data in our backtester to ensure it matches the live production environment. Is this infrastructure parity worth the engineering effort?"

**Akshay Tak:** "It is the single most critical requirement for deployment. The 'Research-Production Gap'—where code works in a backtest but fails in live trading due to data discrepancies—is a major failure mode. Using the same streaming engine for both environments guarantees that the signal seen in the test matches the signal in the market. That reliability is what allows a quant to trust the backtest results."

## 2.6 Interview 6: Developer Experience & System Extensibility

**Interviewee:** Rohan Jha, Quant Developer, GoQuant Technologies

**Focus:** System Functionality, Custom Integrations, and Interoperability

**Q1: Team 76:** "Rohan, regarding system interoperability, we bypassed standard REST APIs in favor of the Model Context Protocol (MCP) to standardize how our Orchestrator discovers and executes tools. Functionally, does this abstraction layer add value for a developer integrating our platform?"

**Rohan Jha:** "It significantly reduces integration debt. In a traditional system, if I want to add a new 'Option Pricing' tool, I have to rewrite the API client, update the agent prompt, and re-deploy the orchestrator. With your MCP implementation, the system functionally acts as a 'Universal Client.' I just expose my tool via the protocol, and the Agent dynamically discovers its capabilities and schema. It turns tool integration from a code-heavy refactor into a simple configuration change."

**Q2: Team 76:** "We designed the 'Domain Agents' (e.g., Strategy Agent) to be open wrappers. This allows internal dev teams to inject their own proprietary analytics logic (like a custom volatility surface model) as functional tools, replacing our defaults. Is this 'Bring Your Own Logic' capability essential?"

**Rohan Jha:** "It is the primary functional requirement for any hedge fund. No sophisticated desk uses out-of-the-box indicators; our edge lies in our proprietary math. If your system were a closed functionality box, it would be useless to us. By architecting the agents to execute injected proprietary code as tools, you transform the product from a generic 'Chatbot' into a 'Reasoning Orchestrator' for our specific IP. That functional flexibility is what allows it to live in a production quant stack."

## 2.7 Interview 7: High-Frequency Systems Engineering

**Interviewee:** Krish Agarwal, Quant Developer, Quadeye

**Focus:** Data Engineering, Latency Optimization, and Signal Robustness

**Q1: Team 76:** "Krish, we moved heavy feature calculation (e.g., Rolling Volatility) out of the database and into in-memory streaming windows. Does this 'Features-as-a-Stream' architecture offer a tangible advantage?"

**Krish Agarwal:** "It fundamentally optimizes the critical path. By eliminating database read-latency, the strategy receives data 'pre-calculated.' This decouples signal generation from execution, keeping the strategy code lightweight and significantly minimizing tick-to-trade latency."

**Q2: Team 76:** "We implemented a 'Regime Monitor' to hot-swap linear models (SARIMAX) for Transformers (Chronos-2) when residuals spike. Is this dynamic adaptability robust enough for production?"

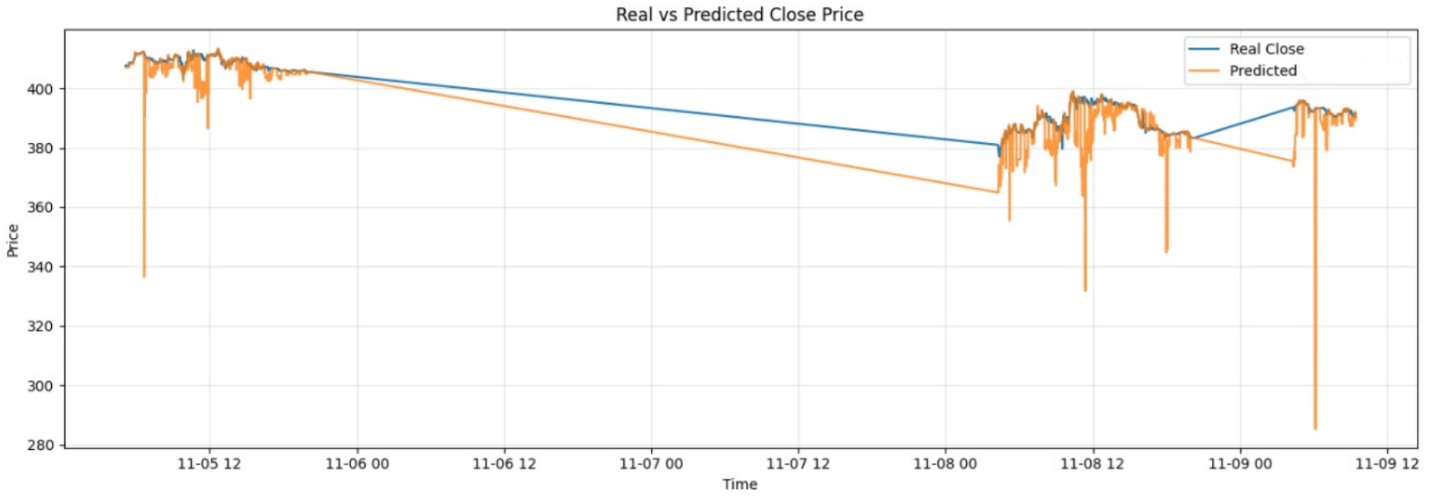
**Krish Agarwal:** "Static models fail in changing regimes. Automating regime detection via residual monitoring is a sophisticated form of risk control. It prevents the system from blindly trading on broken models during volatility, ensuring reliability across varying market conditions."

# 3 Results and Graphical Implementation

## 3.1 Forecasting Performance Visualization

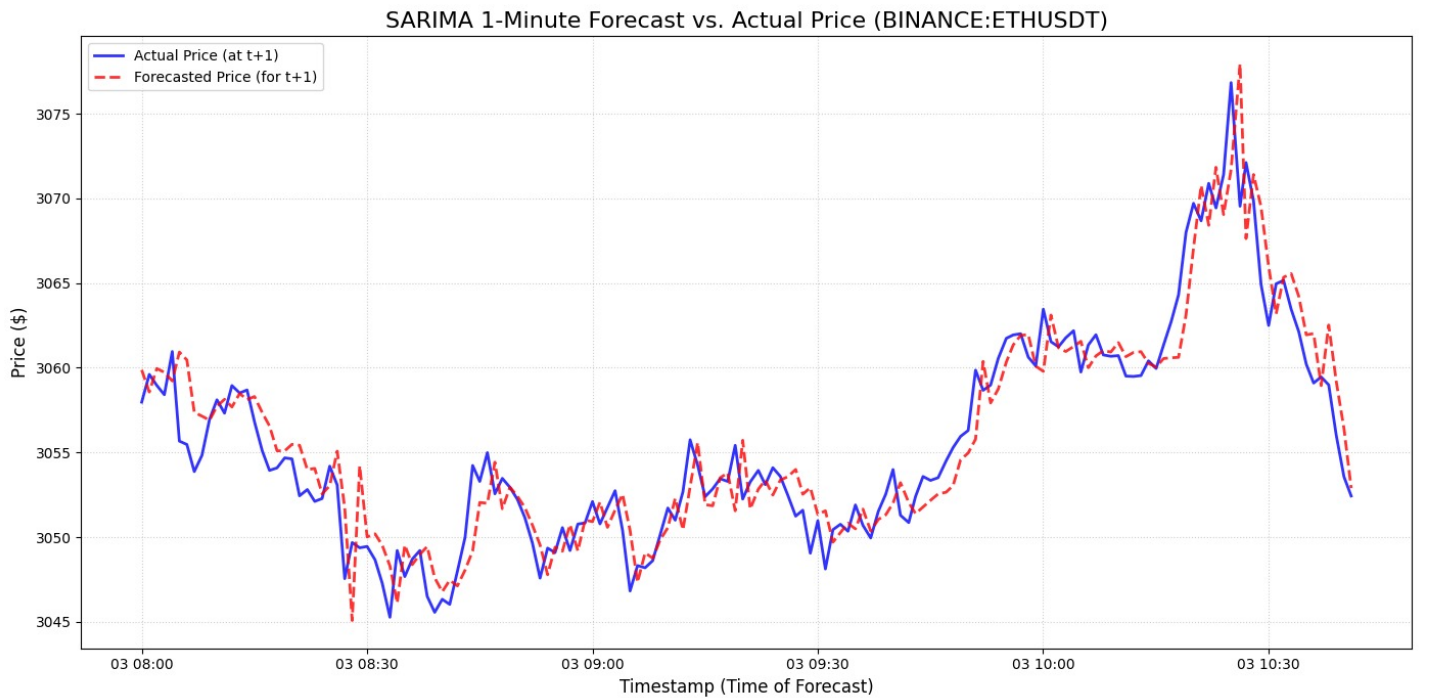
To validate our "Regime-Aware" hypothesis, we visualized the predictive performance of our ensemble models against actual market prices.

Figure 1 illustrates the system's dynamic capability. By leveraging Pathway's stateful stream processing, the engine continuously evaluates the residuals of both the Chronos (Transformer) and SARIMAX (Statistical) models. It automatically switches the active signal source to the model minimizing error in the current regime. This allows the system to capture non-linear volatility using Chronos while reverting to SARIMAX for mean-reverting trends. This modality happens on a horizon of 8.



**Figure 1:** Dynamic Model Switching: Real vs. Predicted Prices (Chronos & SARIMAX Ensemble)

Figure 2 isolates the performance of the statistical baseline. While SARIMAX accurately captures the broad directional trends, the divergence during high-volatility spikes confirms the necessity of our hybrid approach, where deep learning models like Chronos intervene to capture complex, non-stationary behaviors.



**Figure 2:** Statistical Baseline Validation: SARIMAX Forecast vs. Actual Prices

### 3.2 AWS Cloud Infrastructure & Microservices Topology

The platform is engineered as a cloud-native system deployed within a custom AWS Virtual Private Cloud (VPC). The architecture follows a decoupled microservices pattern, prioritizing high-throughput financial modeling and secure, asynchronous data processing. The computational logic is containerized and distributed across Amazon EC2 instances, segregated into functional tiers:

- **Inference & Forecasting:** Compute-intensive modules (e.g., chronos infer, sarimax forecast) operate in isolated subnets to process time-series data.
- **Data Aggregation:** Dedicated microservices for OHLCV, news data, and sentiment scores handle external API ingestion. Network traffic is strictly governed by Public and Private Subnet configurations. Outbound connectivity for private resources is routed via NAT Gateways, while ingress traffic is managed by an AWS Application Load Balancer (ALB), which performs SSL termination and distributes requests across the availability zones.

- **Tactical Backbone:** The core integration layer utilizes Amazon Managed Streaming for Apache Kafka (MSK). This architecture acts as the central nervous system, decoupling data producers from consumers. It facilitates real-time streaming of market signals and orchestrates the complex workflow between the Agentic System and the calculation engines. This ensures that heavy processing loads do not degrade the latency of the client-facing Django interface.
- **Persistence Data Integrity:** Transactional consistency for user portfolios and rebalancing logic is maintained via a Amazon RDS Cluster, configured for high availability and ACID compliance. Unstructured data, including model artifacts and archival logs, is offloaded to Amazon S3, ensuring scalable and durable object storage.
- **Security Observability Stack:** Security is enforced through a defense-in-depth strategy, utilizing Guardrails for input sanitization and granular Security Groups to restrict inter-service communication. The observability pipeline leverages Prometheus for scraping time-series metrics from microservices and MSK brokers. These metrics are visualized in Grafana dashboards to monitor system health, throughput, and latency, with an automated alerting system configured to dispatch notifications via email for anomaly detection.

## 4 User Interface

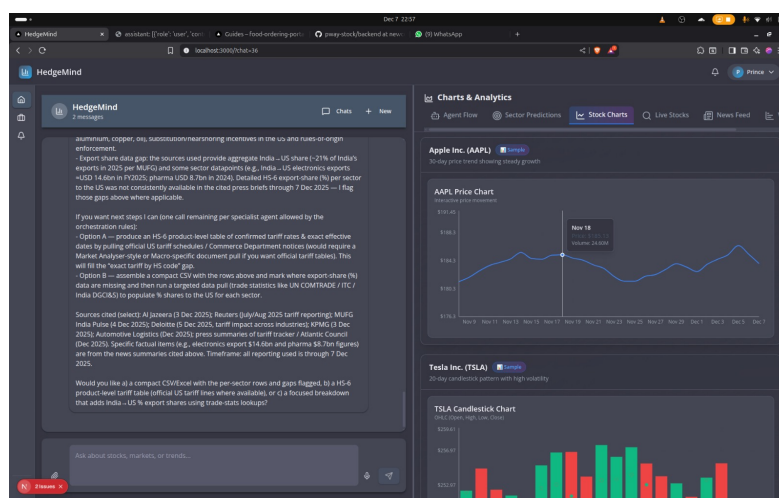


Figure 3: User Interface: AAPL and TSLA Stocks Graphs

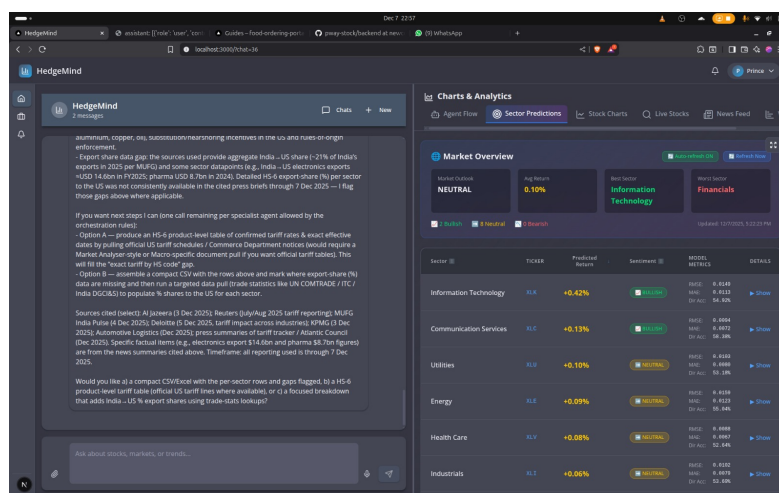


Figure 4: User Interface: Real-Time Sector Predictions

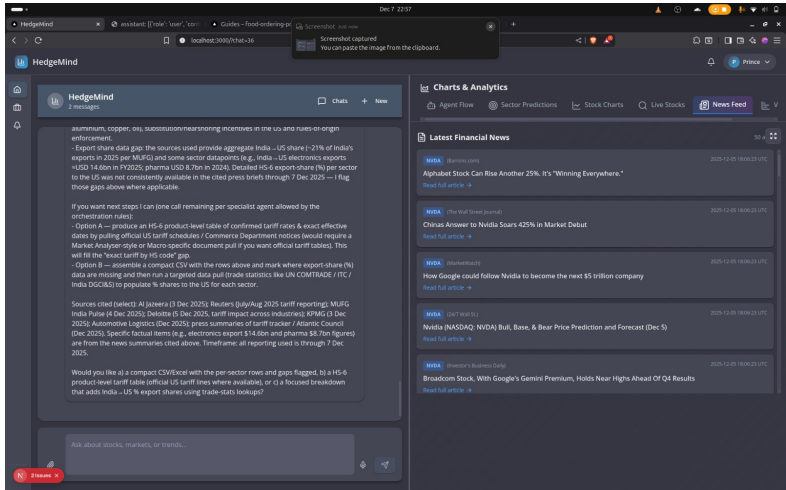


Figure 5: User Interface: Integrated Financial News Feed

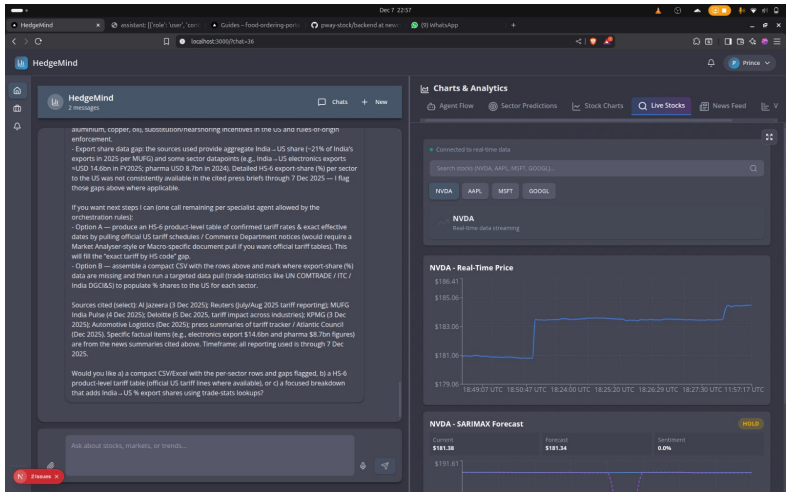


Figure 6: User Interface: Charts and Analysis

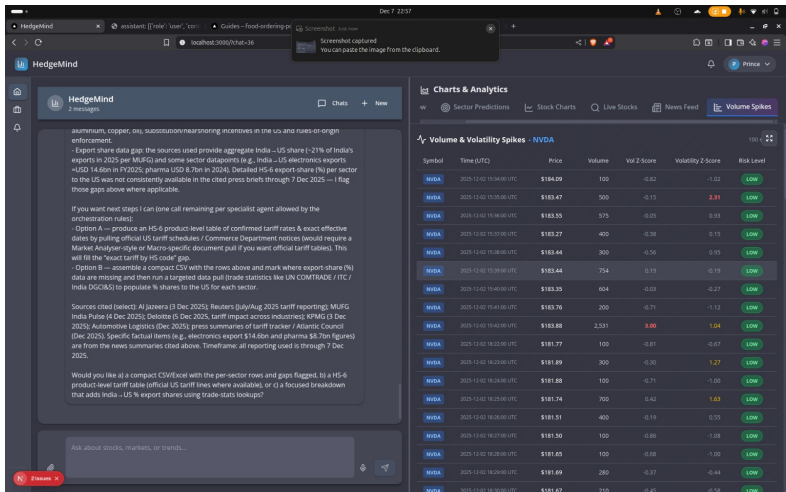


Figure 7: User Interface: Volatility and Volume

# References

[1] Huzaifa Sial. *AI/Personalization Framework*. LinkedIn Pulse, 2025.

[2] Bridgewater Associates. *Research & Insights*. Bridgewater Blog, 2025.

- [3] Bridgewater Associates. *In the News*. Bridgewater Blog, 2025.
- [4] Bridgewater Associates. *The All Weather Story*. Bridgewater PDF, 2025.
- [5] Man Group. *Insights*. Man Group Blog, 2025.
- [6] Man Group. *Hedge Fund Strategy Outlook: Q4 2025: Don't Stop Believin'?*. Man Group Insights, 2025.
- [7] DCF Modeling. *Man Group Limited: History, Ownership, Mission*. DCF Modeling Blog, 2025.
- [8] Tech Blog Writer. *Inside the AI-Powered Hedge Fund: Man Group's Data-Driven Future*. Tech Talks Daily Podcast, 2025.
- [9] Financial Modeling Prep. *Goldman Sachs' "Hedge Fund VIP" Stocks Outperform S&P*. Market News, 2025.
- [10] Goldman Sachs Research. *How to Build a Global Investment Portfolio for the Next Decade*. Goldman Sachs Insights, 2025.
- [11] Goldman Sachs. *Insights*. Goldman Sachs Blog, 2025.
- [12] Day Trading. *Citadel's Trading Strategies [Ken Griffin Trading Philosophy]*. Day Trading Blog, 2025.
- [13] Validea. *How Citadel Claimed The Hedge Fund Crown*. Validea Blog, 2023.
- [14] Citadel. *News*. Citadel Blog, 2025.
- [15] Morgan Stanley. *Hedge Funds: 2024 Outlook and Opportunities*. Morgan Stanley Blog, 2024.
- [16] Morgan Stanley. *Hedge Funds: Investment Products Insights*. Morgan Stanley Blog, 2025.
- [17] Amazon Science. *Chronos-2: From Univariate to Universal Forecasting*. Amazon Science Blog, 2025.
- [18] Anonymous. *Chronos-2: From Univariate to Universal Forecasting*. arXiv preprint arXiv:2510.15821, 2025.
- [19] DataScientest. *SARIMAX model: What is it? How can it be applied to time series*. DataScientest Blog, 2024.
- [20] GeeksforGeeks. *Complete Guide To SARIMAX in Python*. 2023.
- [21] Moonlight.io. *Evolving Financial Trading Strategies with Vectorial Genetic Programming*. Moonlight Literature Review, 2025.
- [22] Anonymous. *Evolving Financial Trading Strategies with Vectorial Genetic Programming*. arXiv preprint arXiv:2504.05418v1, 2025.
- [23] *Lstm-Based Dynamic Correlation Forecasting with Economic Indicators*. SSRN, 2025.
- [24] *Application of Improved LSTM Algorithm in Macroeconomic Forecasting*. NIH PMC, 2021.
- [25] *Enhancing Financial RAG with Agentic AI and Multi-HyDE*. arXiv preprint arXiv:2509.16369v1, 2025.
- [26] *Fin-Rag: A Rag System for Financial Documents*. IJISRT, 2025.
- [27] AI Street. *Man Group Builds Internal AlphaGPT*. AI Street Blog, 2025.
- [28] Man Group. *What AI Can (and Can't Yet) Do for Alpha*. Man Group Insights, 2025.
- [29] Bridgewater Associates. *50 Years of the Bridgewater Daily Observations*. Bridgewater Blog, 2000.
- [30] Wikipedia. *Bridgewater Associates*. 2005.
- [31] Pathway Official Documentation. *Pathway Kafka Integration for Financial Data*. 2025.
- [32] Pathway Official Documentation. *Pathway Model Context Protocol*. 2025.
- [33] Pathway Official Documentation. *Pathway Persistence Engine*. 2025.
- [34] Pathway Official Documentation. *Pathway LLM xPack*. 2025.